Debye Theory of Specific Heat

To resolve the low-temperature failure of Einstein's model—which predicted an exponential drop-off ($C_v \propto e^{-\Theta_E/T}$) that contradicted experimental observations—Peter Debye (1912) recognized that a solid supports long-wavelength collective acoustic oscillations (phonons) whose frequencies extend continuously down to zero ($\omega \to 0$).

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A. Low-Momentum Phonon Dispersion and Density of States

In the long-wavelength / low-momentum limit ($k \to 0$), the acoustic phonon dispersion relation is isotropic and spherically symmetric, simplifying to a linear continuum form:

$$\omega(k) \approx v_s k \implies k = \frac{\omega}{v_s} \tag{1}$$

where v_s is the effective speed of sound in the solid.

In a three-dimensional macroscopic crystal of volume $V = L^3$ with periodic boundary conditions, the allowed wavevector components are quantized as:

$$k_x = \frac{2\pi n_x}{L}, \quad k_y = \frac{2\pi n_y}{L}, \quad k_z = \frac{2\pi n_z}{L} \qquad (n_x, n_y, n_z \in \mathbb{Z}) \tag{2}$$

Due to the **spherical symmetry of the dispersion relation** $\omega(\mathbf{k}) = v_s k$, all states with the same magnitude $k = |\mathbf{k}|$ possess identical energy $\hbar\omega$. Thus, the total number of spatial quantum states within a spherical shell in k-space of radius k and thickness dk is obtained by summing over quantum state numbers:

$$\sum_{n_x,n_y,n_z} = g(k)dk = \frac{V}{(2\pi)^3}\left(4\pi k^2 dk\right) = \frac{Vk^2}{2\pi^2}dk \tag{3}$$

Accounting for 3 polarization modes (1 longitudinal, 2 transverse) for each wavevector k, the density of states $D(\omega)d\omega = 3g(k)dk$ in frequency space becomes:

$$D(\omega)d\omega = 3 \times \frac{V}{2\pi^2}\left(\frac{\omega}{v_s}\right)^2 \frac{d\omega}{v_s} = \frac{3V\omega^2}{2\pi^2 v_s^3}d\omega \tag{4}$$

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B. Energy Calculation Without Debye Cutoff (Infinite Frequency Limit)

Using the Bose-Einstein distribution $\langle n(\omega)\rangle = \frac{1}{e^{\hbar\omega/k_BT}-1}$, the total thermal internal energy U of a 3D harmonic solid without a frequency cutoff ($\omega \in [0, \infty)$) is:

$$U_{\text{uncut}} = \int_0^\infty \hbar\omega\langle n(\omega)\rangle D(\omega)d\omega = \frac{3V\hbar}{2\pi^2 v_s^3}\int_0^\infty \frac{\omega^3}{e^{\hbar\omega/k_BT}-1}d\omega \tag{5}$$

Substituting $x = \frac{\hbar\omega}{k_BT} \implies \omega = \frac{k_BT}{\hbar}x$:

$$U_{\text{uncut}} = \frac{3V\hbar}{2\pi^2 v_s^3}\left(\frac{k_BT}{\hbar}\right)^4 \int_0^\infty \frac{x^3}{e^x - 1}dx \tag{6}$$

Using the standard integral result $\int_0^\infty \frac{x^3}{e^x-1}dx = \frac{\pi^4}{15}$:

$$U_{\text{uncut}} = \frac{\pi^2 V (k_B T)^4}{10\hbar^3 v_s^3} \tag{7}$$

Differentiating with respect to temperature T yields the heat capacity C_v:

$$C_v^{\text{uncut}} = \frac{\partial U_{\text{uncut}}}{\partial T} = \frac{2\pi^2 V k_B^4}{5\hbar^3 v_s^3} T^3 \tag{8}$$

Analysis of the Uncutoff Result

1. **Low Temperature ($T \to 0$):** $C_v \propto T^3$, which **correctly matches experimental observations**.
2. **High Temperature ($T \to \infty$):** $C_v \propto T^3$ grows infinitely without bound, **failing completely** to saturate at the classical Dulong-Petit limit ($C_v \to 3Nk_B$). This non-physical divergence occurs because an infinite continuous medium allows infinitely many high-frequency modes ($\omega \to \infty$).

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C. Introduction of the Debye Cutoff Frequency (ω_D)

Because a real crystal consists of a finite number N of discrete atoms, it possesses a finite total number $3N$ of vibrational degrees of freedom. Debye imposed a high-frequency cutoff ω_D such that the total number of modes integrated over all frequencies equals $3N$:

$$\int_0^{\omega_D} D(\omega)d\omega = 3N \implies \frac{3V}{2\pi^2 v_s^3}\int_0^{\omega_D} \omega^2 d\omega = 3N \tag{9}$$

Evaluating the integral yields:

$$\frac{3V}{2\pi^2 v_s^3}\left(\frac{\omega_D^3}{3}\right) = 3N \implies \omega_D = v_s\left(\frac{6\pi^2 N}{V}\right)^{1/3} \tag{10}$$

We define the **Debye Temperature** Θ_D as:

$$k_B\Theta_D \equiv \hbar\omega_D \implies \Theta_D = \frac{\hbar v_s}{k_B}\left(\frac{6\pi^2 N}{V}\right)^{1/3} \tag{11}$$

Expressing the density of states $D(\omega)$ in terms of N and ω_D:

$$D(\omega) = \begin{cases} \frac{9N}{\omega_D^3}\omega^2, & \omega \le \omega_D \\ 0, & \omega > \omega_D \end{cases} \tag{12}$$

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D. Total Energy and Specific Heat in the Debye Model

With the cutoff frequency ω_D, the internal energy U becomes:

$$U = \int_0^{\omega_D} \hbar\omega \left(\frac{1}{e^{\hbar\omega/k_BT} - 1}\right) \left(\frac{9N}{\omega_D^3}\omega^2\right) d\omega = \frac{9N\hbar}{\omega_D^3} \int_0^{\omega_D} \frac{\omega^3}{e^{\hbar\omega/k_BT} - 1} d\omega \tag{13}$$

Introducing the dimensionless variable $x = \frac{\hbar\omega}{k_BT}$ and upper bound $x_D \equiv \frac{\hbar\omega_D}{k_BT} = \frac{\Theta_D}{T}$:

$$U = 9Nk_BT \left(\frac{T}{\Theta_D}\right)^3 \int_0^{\Theta_D/T} \frac{x^3}{e^x - 1} dx \tag{14}$$

Differentiating U with respect to T yields the **Debye Specific Heat**:

$$C_v = \left(\frac{\partial U}{\partial T}\right)_V = 9Nk_B \left(\frac{T}{\Theta_D}\right)^3 \int_0^{\Theta_D/T} \frac{x^4 e^x}{(e^x - 1)^2} dx \tag{15}$$

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E. Temperature Limits of the Full Debye Model

1. Low-Temperature Limit ($T \ll \Theta_D$)

When $T \to 0$, the upper integration limit goes to infinity ($x_D = \frac{\Theta_D}{T} \to \infty$). Replacing the integral with its definite value $\int_0^\infty \frac{x^3}{e^x - 1} dx = \frac{\pi^4}{15}$:

$$U \approx 9Nk_BT \left(\frac{T}{\Theta_D}\right)^3 \left(\frac{\pi^4}{15}\right) = \frac{3\pi^4}{5} Nk_B \frac{T^4}{\Theta_D^3} \tag{16}$$

Taking the derivative with respect to T gives the **Debye T^3 Law**:

$$C_v \approx \frac{12\pi^4}{5} Nk_B \left(\frac{T}{\Theta_D}\right)^3 \propto T^3 \tag{17}$$

Result: Successfully recovers the experimental T^3 power law for low-temperature lattice heat capacity.

2. High-Temperature Limit ($T \gg \Theta_D$)

When $T \to \infty$, $x \ll 1$ throughout the integral domain. Expanding $e^x \approx 1 + x$:

$$U \approx 9Nk_BT \left(\frac{T}{\Theta_D}\right)^3 \int_0^{\Theta_D/T} \frac{x^3}{(1 + x) - 1} dx = 9Nk_BT \left(\frac{T}{\Theta_D}\right)^3 \int_0^{\Theta_D/T} x^2 dx \tag{18}$$

$$U \approx 9Nk_BT \left(\frac{T}{\Theta_D}\right)^3 \left[\frac{1}{3}\left(\frac{\Theta_D}{T}\right)^3\right] = 3Nk_BT \tag{19}$$

Differentiating with respect to T:

$$C_v \approx \frac{\partial(3Nk_BT)}{\partial T} = 3Nk_B \tag{20}$$

Result: With the introduction of the cutoff ω_D, Debye's model successfully saturates at high temperatures to recover the classical **Dulong–Petit Law** ($C_v = 3Nk_B$).

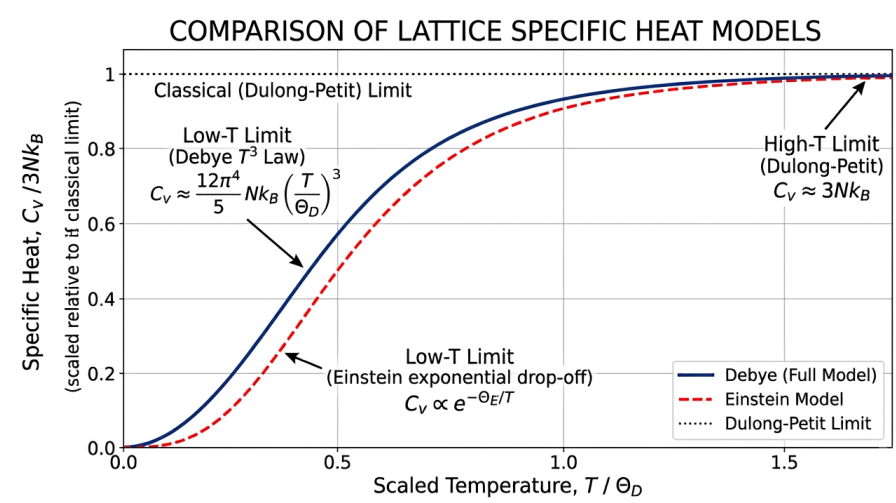


Figure 1: Temperature dependence of lattice specific heat $C_v/3Nk_B$ as a function of T/Θ_D. The curve demonstrates the T^3 power law behavior at $T \ll \Theta_D$ and smooth asymptotic approach toward the Dulong–Petit classical limit ($3Nk_B$) at high temperatures $T \gg \Theta_D$.